

Quantitative Volumetric Analysis of Retinal Ischemia with an Oxygen Diffusion Model and OCT Angiography

Pengxiao Zang, PhD,¹ Tristan T. Hormel, PhD,¹ Thomas S. Hwang, MD,¹ Yali Jia, PhD^{1,2}

Purpose: Retinal ischemia is a major feature of diabetic retinopathy (DR). Traditional nonperfused areas measured by OCT angiography (OCTA) measure blood supply but not ischemia. We propose a novel 3-dimensional (3D) quantitative method to derive ischemia measurements from OCTA data.

Design: Cross-sectional study.

Participants: We acquired 223 macular OCTA volumes from 33 healthy eyes, 33 diabetic eyes without retinopathy, 7 eyes with nonreferable DR, 17 eyes with referable but nonvision-threatening DR, and 133 eyes with vision-threatening DR.

Methods: Each eye was scanned using a spectral-domain OCTA system (Avanti RTVue-XR, Visionix/Optovue, Inc) with 1.6-mm scan depth in a 3 × 3-mm region (640 × 304 × 304 voxels) centered on the fovea. For each scanned OCTA volume, a custom algorithm removed flow projection artifacts. We then enhanced, binarized, and skeletonized the vasculature in each OCTA volume and generated a 3D oxygen tension map using a zero-order kinetics oxygen diffusion model. Each volume was scaled to the average retina thickness in healthy controls after foveal registration and flattening of the Bruch's membrane. Finally, we extracted 3D ischemia maps by comparison with a reference map established from scans of healthy eyes using the same processing. To assess the ability of the ischemia maps to grade DR severity, we constructed receiver operating characteristic curves for diagnosing diabetes, referable DR, and vision-threatening DR.

Main Outcome Measures: Spearman correlation coefficient and area under receiver operating characteristic curve (AUC) were used to quantify the ability of the ischemia maps to DR.

Results: The ischemia maps showed that the ischemic tissues were at or near pathologically nonperfused areas, but not the normally nonvascular tissue, such as the foveal avascular zone. We found multiple novel metrics, including inferred 3D-oxygen tension, ischemia index, and ischemic volume ratio, were strongly correlated with DR severity. The AUCs of ischemia index measured were 0.94 for diabetes, 0.89 for DR, 0.88 for referable DR, and 0.85 for vision-threatening DR.

Conclusions: A quantitative method to infer 3D oxygen tension and ischemia using OCTA in diabetic eyes can identify ischemic tissue that are more specific to pathologic changes in DR.

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Supplemental material available at www.ophtalmologyscience.org.

Diabetic retinopathy (DR) is a leading cause of preventable blindness globally.¹ Early detection of this disease is crucial for successful treatment outcomes.^{2–4} Retinal ischemia is a key feature of DR that drives the downstream complications, making it an important pathology to characterize.^{5,6} The current OCT angiography (OCTA)-based methods to assess ischemia have focused on vessel density and nonperfusion areas (NPAs), both of which measure the absence of vessels without reference to normal.^{7–10} However, nonperfusion can be physiologic, as in the foveal avascular zone (FAZ).

More importantly, these measures of nonperfusion do not directly measure ischemia, which implies inadequate blood flow of any given tissue. A true measure of ischemia would require the knowledge of metabolic demand and diffusion of oxygen at a point. Without observing frank tissue abnormalities, such as opacification of the retina in arterial occlusions, it is impossible to measure the imbalance between the metabolic demand and oxygen supply. Currently, the direct measurements of retinal tissue oxygen and ischemia are mostly based on microelectrodes and fluorescence lifetime imaging ophthalmoscopy which can

measure oxygen concentration or partial oxygen pressure on animal models.^{11,12} But they could not be used on living human eyes. In addition, the visible light OCT can generate optical imaging of oximetry and has a potential of working on living human eyes. But the generated optical imaging is only the measurement of vascular oximetry, not the oxygen tension within the retinal tissue.¹³ It is worth noting that one can estimate normal perfusion at any given point of the retina based on the spatial relationship between tissue and blood vessels in the normal retina. By comparing the spatial relationship between vessels and tissue against the reference created by averaging of the retina from normal eyes, we propose a model that indirectly measures ischemia, not just nonperfusion. We hypothesize that this method will more reliably detect ischemic changes in DR compared with measurements of nonperfusion, while avoiding detection of pathology in normally avascular areas, such as the FAZ.

To obtain an indirect estimate of ischemic regions, volumetric anatomic information should be supplemented with oxygen diffusion dynamics in order to better infer tissue oxygen saturation. Therefore, to detect ischemia, we propose combining vascular anatomic measurements from OCTA with an oxygen diffusion model.

Methods

Data Acquisition

We included healthy controls and diabetic patients with DR severity ranging from no clinically evident retinopathy to proliferative DR. One or both eyes of each participant underwent 7-field color fundus photography and an OCTA scan using a commercial 70-kHz spectral-domain OCT system (RTVue-XR Avanti, Optovue/Visionix Inc) with 840-nm central wavelength. The scan depth was 1.6 mm in a 3×3 -mm region ($640 \times 304 \times 304$ voxels) centered on the fovea. Two repeated B-frames were captured at each cross-sectional location. Structural images were obtained by averaging the 2 repeated and registered B-frames. Blood flow was detected using the split-spectrum amplitude-decorrelation angiography algorithm.¹⁴ For each volume, 2 continuously acquired raster scans (1 x-fast scan and 1 y-fast scan) were registered and merged with an orthogonal registration algorithm to reduce motion artifacts.¹⁵ In addition, a projection-resolved OCTA algorithm was applied to all OCTA scans to remove flow projection artifacts.¹⁶ Scans with a signal strength index <50 were excluded.

To quantitatively analyze the oxygen tension and ischemia maps across different DR severities, a masked trained retina specialist (T.S.H.) graded 7-field color fundus photographs based on ETDRS scale.^{17,18} The presence of diabetic macular edema (DME) was determined using the central subfield thickness from structural OCT based on the Diabetic Retinopathy Clinical Research Network standard.¹⁹ Diabetic retinopathy severity was defined as nonreferable DR when the ETDRS level was better than 35 and without DME; referable but nonvision-threatening DR when ETDRS level was between 35 and 47 without DME; and vision-threatening DR when ETDRS level 53 or worse or any stage of DR with DME.²⁰ Participants were enrolled after informed consent in accordance with an Institutional Review Board (Oregon Health & Science University) approved protocol. This study complied with the Declaration of Helsinki and the Health Insurance Portability and Accountability Act.

Framework Summary

Our 3-dimensional (3D) oxygen tension and ischemia quantitative analysis framework includes several steps (Fig. 1).

Data Preprocessing

To improve the computational efficiency, a subvolume from the inner limiting membrane to the bottom of Bruch's membrane plus 20 voxels (including choriocapillaris) was extracted from each original OCTA volume. In preprocessing, the vasculature in both the retina and choroid of each subvolume was enhanced using the optimally oriented flux method.^{21,22} The enhanced vasculature was then binarized using an empirically selected threshold of the mean plus 0.1 standard deviations of the nonzero voxels. The binary images were then skeletonized using a medial axis transform algorithm.^{23,24}

Oxygen Diffusion Calculation

An oxygen diffusion model with zero-order kinetics (Fig. 2) was used to generate a convolutional kernel K to apply to the skeletonized OCTA volumes in order to generate an oxygen tension map²⁵ according to

$$K = \begin{cases} \left(1 - \frac{d}{L_p}\right)^2 & \text{for } 0 \leq d \leq L_p \\ 0 & \text{for } x > L_p \end{cases} \quad (1)$$

where d is the distance from the voxel being acted on by the kernel and L_p is the penetration depth given by

$$L_p = \sqrt{2D_t\alpha_t P/M} \quad (2)$$

where D_t is the oxygen diffusion coefficient 1.97×10^{-5} cm²/s in tissue,²⁶ α_t is the solubility coefficient 2.4×10^{-5} mL O₂/mL tissue/mmHg,²⁷ P is oxygen tension in the target vessel,¹¹ and M is retinal tissue oxygen consumption.¹¹ To get a L_p close to the real value, P and M were respectively selected as the oxygen tension of retinal arteries 101 mmHg and average retinal tissue oxygen consumption 1.8 ml O₂/100 g tissue/min measured from Long-Evans rat in light adaptation.¹¹ In addition, we have assumed equal oxygen tension within the vessels and homogenous consumption throughout the retina.

To transfer L_p from millimeters to voxels, each skeletonized OCTA volume was resized to isotropic resolution before the oxygen diffusion calculation. Oxygen tension maps T are then obtained by convolving the established kernels with the skeletonized vasculatures

$$T = V * K \quad (3)$$

where $*$ is the convolution operation, V are the set of vascular voxels, and K was the generated convolutional kernel.

Retina Rescaling

After the generation of oxygen tension volume, the thickness of each retinal layer in the volume was rescaled based on a reference thickness map. To obtain the reference thickness map, we first calculated a thickness map based on the commercial software in the spectral-domain OCT system in this study (RTVue-XR Avanti, Optovue/Visionix Inc). In addition, for the cases with severe pathologies, the automated layer segmentation was manually corrected by graders using our COOL-ART software.^{28,29} The following retinal layers or boundaries were involved in the generation of the thickness map: the inner limiting membrane, nerve fiber layer, ganglion cell layer, inner plexiform layer, inner nuclear layer, outer plexiform layer, outer nuclear layer, ellipsoid

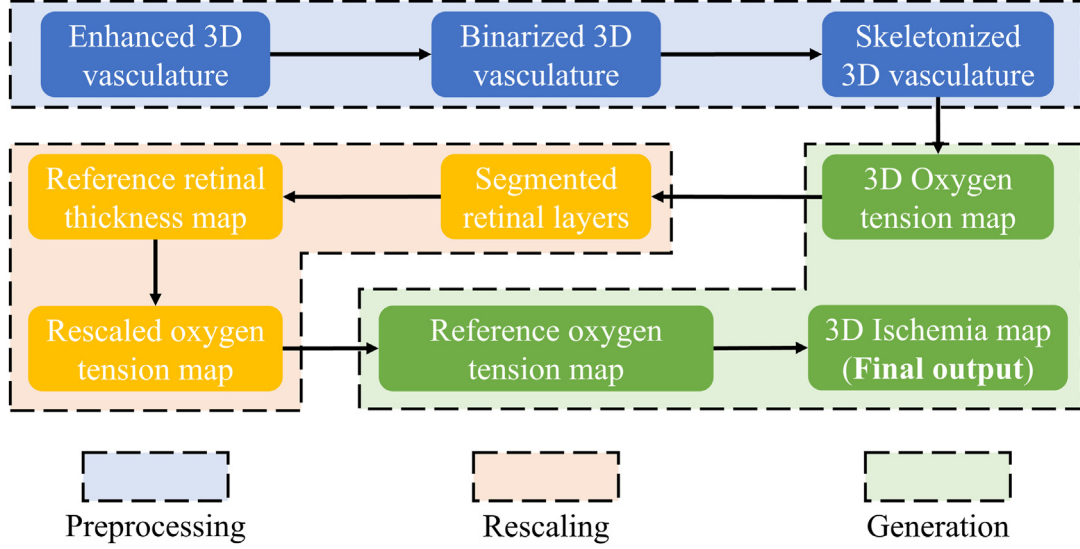


Fig. 1. The framework of the proposed 3D quantitative method for oxygen tension and ischemia of OCTA volume. In preprocessing (blue boxes), we first volumetrically enhanced, binarized, and skeletonized the vasculature. The oxygen tension map was then calculated based on the skeletonized volume using an oxygen diffusion model (green box on the second line). We next rescaled each oxygen tension map based on a reference thickness map, which was the average of all healthy controls (yellow boxes). Finally, the ischemia map was generated by comparison with a reference oxygen tension map established by combining half of the healthy volumes (bottom 2 green boxes). 3D = 3-dimensional; OCTA = OCT angiography.

zone, retinal pigment epithelium, and Bruch's membrane (Fig. 3). A reference thickness map was constructed by averaging the thickness of each retinal layer of all the healthy cases after foveal registration and Bruch's membrane flattening. To obtain the rescaled volumes, we registered individual scans to the reference thickness map by centering each on the fovea. The fovea was detected as the center of the binary mask of the FAZ, which was segmented using an empirically selected threshold on the combined image of inner retinal en face OCTA projection and thickness map. Each retinal layer was then rescaled using linear interpolation based on the thickness of the corresponding layer in the reference thickness map (Fig. 3).

Ischemia Map Generation

After all the oxygen tension maps were rescaled, all the healthy maps were split to a development set with 17 volumes and a testing set with 16 volumes. Oxygen tension maps $T_{\min}$ and T_{σ} were then respectively generated as the minimum and mean minus 2.5

times standard deviation among all the maps in the development set (4). The reference map T_{ref} was then generated as the maximum between $T_{\min}$ and T_{σ} . The ischemia map of each OCTA volume could be finally generated as

$$T_{\text{ref}} = T_{\min} - 2.5 \times T_{\sigma}$$

$$M = \frac{\text{Max}(T_{\text{ref}} - T, 0)}{T_{\text{ref}}} \quad (4)$$

where M is the generated ischemia map, and T is the oxygen tension map of the corresponding case. In addition, the testing set and maps of other DR severities were only used to evaluate the generated reference oxygen tension map.

Ischemia Quantification

Before the quantification of ischemia, it was important to clarify that the voxel values in the generated oxygen tension maps could only estimate the relative differences in the distribution of oxygen tension and do not represent the actual oxygen tension values in the

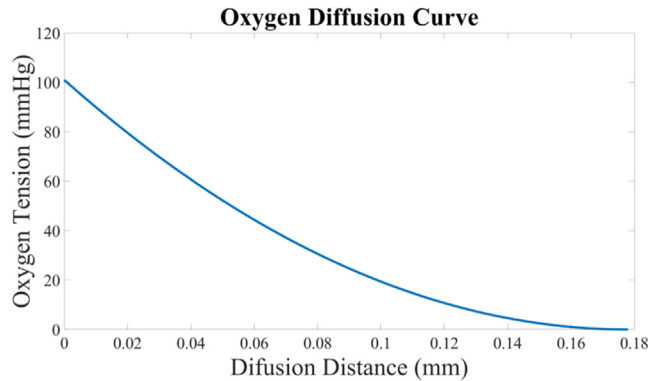


Fig. 2. The oxygen diffusion curve for vessels in the retina and choroidal using the parameters described in the main text.

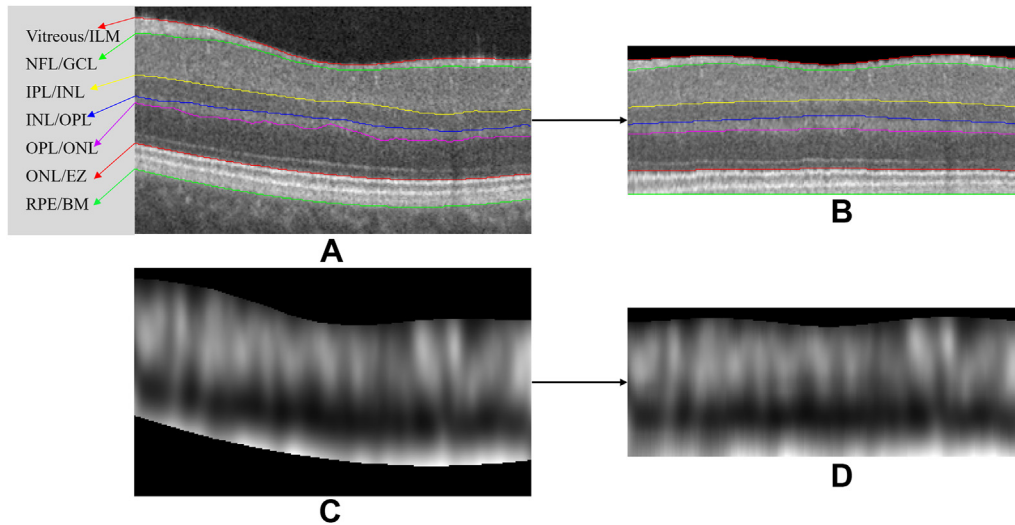


Fig. 3. Rescaling based on retinal thickness to obtain a reference volume. For each retinal layer, each A-line was rescaled using linear interpolation based on the thickness of the same layer at the same en face position. **A**, The segmented 6 retinal layers based on the 7 marked boundaries in the original B-scan before rescaling. **B**, The rescaled and flattened (based on RPE/BM) B-scan from (A). Each retinal layer was rescaled to the exact same shape of the reference thickness map. **C**, The B-scan oxygen tension with the same position of (A). **D**, The rescaled B-scan oxygen tension from (C) using the rescaling shown from (A) to (B). BM = Bruch's membrane; EZ = ellipsoid zone; GCL = ganglion cell layer; ILM = inner limiting membrane; INL = inner nuclear layer; IPL = inner plexiform layer; NFL = nerve fiber layer; OPL = outer plexiform layer; ONL = outer nuclear layer; RPE = retinal pigment epithelium.

corresponding eye. To ensure that the estimated oxygen tensions fall within the actual oxygen tension values range, they were normalized by dividing them by the target vessel oxygen tension $P = 101$ mmHg.¹¹ To quantitatively analyze the generated ischemia map which was generated based on the normalized oxygen tension map, we defined 2 ischemia metrics. The first metric was the ischemia index, defined as the percentage of oxygen tension loss compared with the oxygen tension in the reference map (calculated using Eq. (4)). The ischemia index could be calculated for each voxel in the ischemia map. The second metric was the ischemic volume ratio, defined as the percentage of ischemic voxels compared with the total volume of the retina. The ischemic volume ratio could be calculated for each volumetric ischemia map.

Evaluation on Different Retinal Layers

To analyze the 3D oxygen tension and ischemia maps on different retinal layers, 3 retinal capillary plexus layers, the superficial vascular complex (SVC), intermediate capillary plexus (ICP), and deep capillary plexus (DCP), were selected.^{7,30,31} The SVC was defined as the inner 80% of the ganglion cell complex, which included all structures between the inner limiting membrane and inner plexiform layer/inner nuclear layer border. The ICP was defined as the outer 20% of the ganglion cell complex and the

inner 50% of the inner nuclear layer. The DCP was defined as the remaining slab internal to the outer boundary of the outer plexiform layer (Fig. 3).²⁹ The average en face projections of the 3D oxygen tension and ischemia maps were generated on 3 selected retinal layers for both qualitative and quantitative analysis. Additionally, the generated en face projections could be also used to compare with traditional nonperfusion maps which were generated on the same retinal layers based on previously developed models in our group.^{32–34}

Results

We recruited and examined 20 healthy participants and 179 patients with diabetes at the Casey Eye Institute, Oregon Health & Science University in the United States. The data characteristics are shown below (Table 1).

Qualitative Analysis

Compared with the segmented NPA, which focused on the capillary dropout,³⁰ our ischemia map attempts to focus on retinal tissues that lacked oxygen supply by modeling diffusion dynamics. We firstly compare the generated 3D

Table 1. Data Distribution and Characteristic

Severity	Healthy	nDR	nrDR	r_nvDR	vtDR
Volume number	33	33	7	17	133
Age, mean (SD), yrs	37.9 (12.3)	54.1 (15.7)	45.3 (15.1)	60.5 (11.3)	58.7 (12.2)
Female, %	54.8%	48.5%	71.4%	64.7%	51.1%
SSI, mean (SD)	78.2 (7.7)	69.9 (7.5)	75.7 (7.3)	68.5 (7.7)	66.4 (7.1)

nDR = diabetic without diabetic retinopathy; nrDR = nonreferable diabetic retinopathy; r_nvDR = referable but nonvision-threatening diabetic retinopathy; SD = standard deviation; SSI = signal strength index; vtDR = vision-threatening diabetic retinopathy.

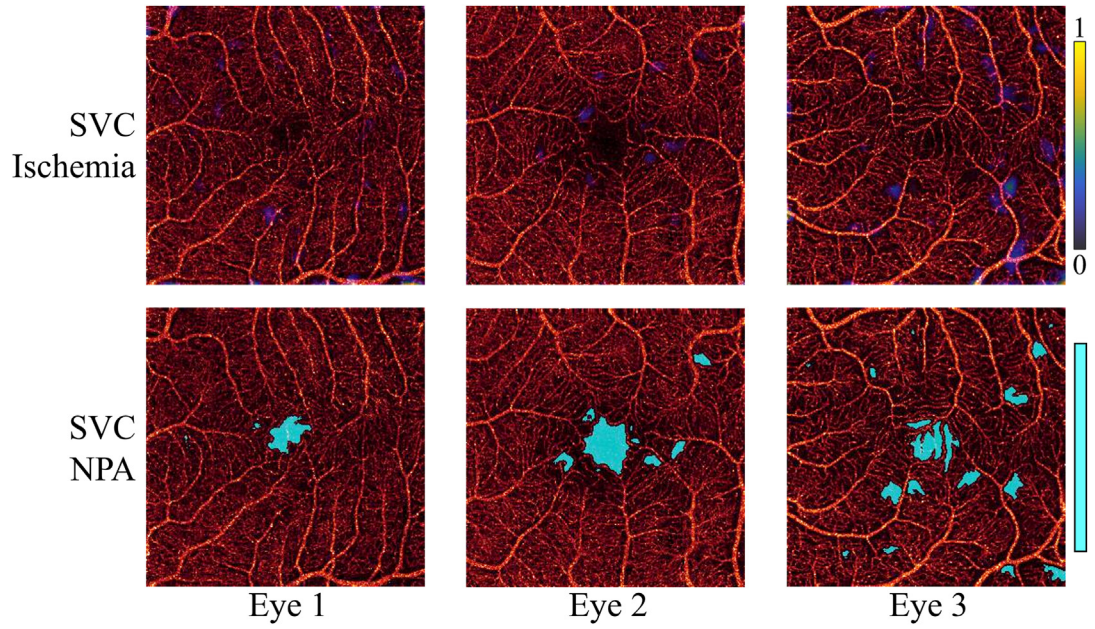


Fig. 4. Generated 3D ischemia map of 3 healthy eyes in the testing set. The mean projection of SVC from each ischemia map was calculated and overlaid on the corresponding OCTA en face image. Compared with the segmented NPA,³² the ischemia map did not identify every region that lacked vessels, instead highlighting regions with poor inferred oxygen concentration. That the ischemia map avoided detecting nonpathological capillary drop out (e.g., in the foveal avascular zone) suggests that it may be a more sensitive indicator of pathology than NPA measurements. 3D = 3-dimensional; NPA = nonperfusion area; OCTA = OCT angiography; SVC = superficial vascular complex.

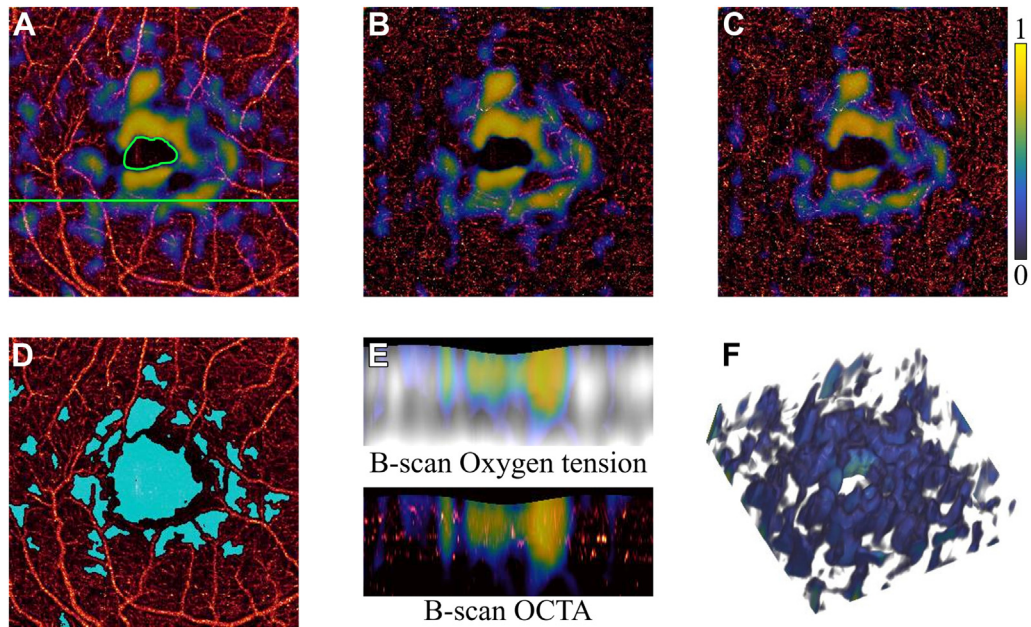


Fig. 5. Generated 3D ischemia map of a vision-threatening diabetic retinopathy OCTA volume, showing the ischemia distribution in different retinal layers. The ischemic areas broadly match the nonperfusion area, but avoid the FAZ. **A**, En face SVC projection of the ischemia map overlaid on an en face OCTA image of the SVC. The healthy avascular tissue was demarcated with a green boundary. **B**, En face intermediate capillary plexus projection of the ischemia map. **C**, En face deep capillary plexus projection of the ischemia map. **D**, Segmented nonperfusion area overlaid on the SVC.³² The nonperfusion areas appear similar to the ischemic areas, with the exception of the FAZ. **E**, A B-scan ischemia map overlaid on the B-scan oxygen tension and OCTA image marked by a green line in (A). **F**, 3D view of the ischemia volumes. 3D = 3-dimensional; FAZ = foveal avascular zone; OCTA = OCT angiography; SVC = superficial vascular complex.

Table 2. The Correlations between 3 Ischemia Metrics (Mean $\pm$ SD) and DR Severities

Mean $\pm$ SD	Numbers	Oxygen Tension	Ischemic Volume Ratio	Ischemia Index
Healthy	16	56.75 $\pm$ 2.48	8.56% $\pm$ 3.96%	1.12% $\pm$ 0.65%
nDR	33	52.86 $\pm$ 3.43	16.99% $\pm$ 7.87%	2.73% $\pm$ 2.09%
nrDR	7	50.43 $\pm$ 5.04	23.55% $\pm$ 12.01%	3.71% $\pm$ 2.99%
r_nvtDR	17	50.07 $\pm$ 5.27	26.60% $\pm$ 12.82%	4.99% $\pm$ 3.31%
vtDR	133	47.63 $\pm$ 5.64	33.77% $\pm$ 13.20%	7.96% $\pm$ 5.02%
Spearman correlation coefficient along severities (<i>P</i> values)		-0.49*	0.58*	0.50*

DR = diabetic retinopathy; nDR = diabetic without diabetic retinopathy; nrDR = nonreferable diabetic retinopathy; r_nvtDR = referable but nonvision-threatening diabetic retinopathy; SD = standard deviation; vtDR = vision-threatening diabetic retinopathy.

**P* value < 0.001.

ischemia maps with segmented NPA maps in 3 healthy eyes. Notably, the en face projections of the 3D ischemia maps did not highlight the FAZ, which, in contrast, was segmented as NPAs (Fig. 4). Additionally, the ischemia maps highlighted much smaller areas compared with the NPA maps in these 3 healthy eyes. On the other hand, the ischemia map of a vision-threatening DR eye clearly shows 3D ischemic regions (Fig. 5). In this example, the highlighted ischemic regions appeared similar to the pathological NPAs (Fig. 5D). By overlaying the B-scan oxygen tension map and OCTA (Fig. 5E), we could examine how the ischemia volumes were distributed in cross-sectional view; therefore, the ischemic regions can be revealed volumetrically (Fig. 5F).

Quantitative Analysis

To analyze the correlations between the generated 3D ischemia maps and DR severity, we calculated the average oxygen tension, average ischemia index, and ischemic volume ratio for the OCTA volumes in each DR severity (Table 2, Fig. 6). For the 5 DR severities examined, lower oxygen tension, higher ischemia index, and ischemic volume ratio showed strong correlations with increased DR severity (Table 2; Spearman correlation coefficients). In addition, compared with the ischemic volume ratio, the ischemia index achieved a higher area under the receiver

operating characteristic curve (AUC) for diagnosing DR, referable DR, and vision-threatening DR (Table 3).

Plexus-Specific Analysis and the Comparison to NPAs

To evaluate whether ischemia metrics perform better in the plexus-specific slabs, we analyzed Spearman correlation coefficient and AUC for each retinal plexus (Table S4 and S5, available at www.opthalmologyscience.org). We found that the ICP had the highest oxygen tension among all DR severities (Table S4, available at www.opthalmologyscience.org). The SVC had the highest ischemic volume ratio and ischemia index and achieved the highest AUC for diabetes diagnosis compared with ICP and DCP (Table S5, available at www.opthalmologyscience.org). The DCP achieved the highest AUC in discriminating different levels of DR compared with SVC and ICP (Table S5, available at www.opthalmologyscience.org). However, the ischemia biomarkers using the whole retina achieved higher diagnostic performances than individual plexuses in all severities (Table 3).

To verify whether our oxygen tension and ischemia maps correlated with the capillary dropout, the Spearman correlation coefficients between the mean NPAs and oxygen

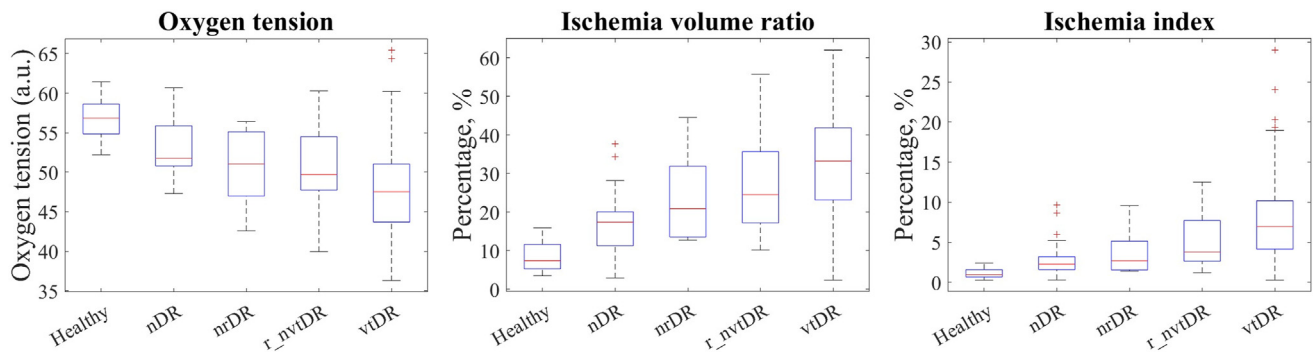


Fig. 6. Boxplots based on average oxygen tensions, ischemic volume ratio, and ischemia index of 5 different DR severities: healthy, diabetic without DR, nonreferable DR, referable but nonvision-threatening DR, and vision-threatening DR. For both ischemic volume ratio and index, the median values (red lines in the middle and right figures) were positively correlated with DR severity. For the oxygen tensions, the median values (red lines in the left figure) were negatively correlated with DR severity. DR = diabetic retinopathy; nDR = diabetic without diabetic retinopathy; nrDR = nonreferable diabetic retinopathy; r_nvtDR = referable but nonvision-threatening diabetic retinopathy; vtDR = vision-threatening diabetic retinopathy.

Table 3. AUCs of 3 Ischemia Metrics on Diagnosing DR with Variant Severities

	Oxygen Tension	Ischemic Volume Ratio	Ischemia Index
Diabetes	0.10	0.95	0.94
DR	0.18	0.89	0.89
rDR	0.19	0.87	0.88
vtDR	0.22	0.83	0.85

AUC = area under receiver operating characteristic curve; DR = diabetic retinopathy; rDR = referable diabetic retinopathy; vtDR = vision-threatening diabetic retinopathy.

The bold values indicate the highest AUC for each task.

tension and ischemia maps were calculated (Table S6, available at www.ophtalmologyscience.org). The mean oxygen tensions and NPAs were negatively correlated. The ischemic volume ratios and mean indexes were positively correlated with the NPA. The high correlations shown in Table S6 (available at www.ophtalmologyscience.org) demonstrate our method could correctly reflect the oxygen tension and ischemia changes due to capillary dropout.

Discussion

In this study, we proposed a 3D quantitative analysis method to estimate oxygen tension and ischemia in OCTA volumes of subjects with different DR severities. We generated oxygen tension and ischemia maps based on 3D microvasculatures in OCTA volumes. Unlike current nonperfusion detection methods, our ischemia maps highlighted only pathologic changes that result in lower oxygen levels while ignoring normal avascular tissue, such as in the FAZ. To the best of our knowledge, our method is the first 3D analysis method for both oxygen tension and ischemia based on OCTA volume.

Our method used an oxygen diffusion model with zero-order kinetics to generate the oxygen tension map based on enhanced and skeletonized vessels. For model parameters, we used values measured from Long-Evans rats in light adaptation in the oxygen diffusion model.¹⁷ While some parameters, such as oxygen tension in vessels or consumption in the retina, likely vary with location, assessing local variation directly is difficult. We instead chose a simpler model to determine ischemic locations by using values from the literature rather than fit values for model parameters. This approach avoids overfitting the data and ensures the results are reasonable. In addition, our model produced reasonable oxygen tension values in the healthy avascular regions; the estimated oxygen tension values in the outer retina within and outside the FAZ were similar.

The other key aspect of our approach is the reference map. To generate this reference map, we performed 2 crucial procedures to make it more objective. First, we used skeletonized vasculatures to prevent the varied distributions of large vessels present in different healthy volumes from influencing the oxygen tension distribution in the reference map. Second, we separated the healthy

controls into a development set and a testing set and only used the development set to establish the reference map. This approach allowed verification of the reliability of our results.

Compared with current OCTA-based nonperfusion analysis methods,^{7–10} our method has 3 major technical advantages. First, our oxygen tension map was generated by combining a physical model with real-world measurements. This contrasts with conventional image processing kernels, which are empirically determined and do not incorporate physiological information. By including measurements from physiologic models, the model is more than a mathematical construct but a physical modeling of oxygen tension. Second, we generated the ischemia map by comparing it with a reference oxygen tension map. As a result, our generated map focused only on the retinal tissues with oxygen levels lower than normal and disregarded the normal avascular tissue (Fig. 4). Finally, our ischemia maps were generated volumetrically, preserving the 3D of interaction between perfusion and tissue in the retina. Unlike NPA segmentation, ischemia should be quantified as a volumetric entity, not in separate plexuses. This could also explain why the ischemia biomarkers using the whole retina achieved higher diagnostic performances compared with individual plexuses in all severities.

These ischemia-based metrics have 2 potential clinical advantages. Because they specifically identify pathologic changes, they may be more accurate when used as a screening tool across population variation. The variation in FAZ size has been widely reported.^{35,36} Less is known about variations in avascular zones near larger vessels. An objective and quantitative metric that is specific to vascular damage from diabetes can potentially improve on the current photography-based screening, and even incremental improvement in accuracy can have a large impact on the efficiency of screening. Secondly, by directing clinician attention to the ischemic, and not physiologically avascular areas, it has the potential of providing clinicians with more accurate insight about the severity and localization of the disease.

This study had several limitations. First, a longitudinal study is needed to establish causality of the proposed model. Second, the model did not consider the differences between arteries and veins, large vessels and microvasculature, and the retina and choroid. It is likely that these differences affect our model parameters. In addition, the reference oxygen tension map was generated only based on 17 healthy controls. In future studies, we plan to use a larger number of eyes scanned from a more diverse population for the reference volume and allow parameters to vary in different vessels. Lastly, the model parameters were measured from animal studies but not human eyes. This shortcoming can be potentially compensated by integrating our current method with a visible light OCTA system to calculate the oxygen tension maps based on the measured oximetry in different vessels, thereby enhancing the accuracy of our oxygen tension map.^{13,37}

We proposed a 3D quantitative analysis method for the oxygen tension and ischemia of eyes with or without DR. The 3D oxygen tension and ischemia maps were generated using an oxygen diffusion model with zero-order kinetics based on skeletonized vasculature in OCTA volume. Lower oxygen

tension, higher ischemia index, and ischemic volume ratio from the model were strongly correlated with increased DR severity. The ischemia maps did not identify normal avascular tissue as ischemic tissue. Our method offers a 3D quantitative

analysis of oxygen tension and retinal ischemia in DR. The generated oxygen tension and ischemic maps could be used in future studies of retinal oxygen supply reduction caused by DR or other eye diseases.

Footnotes and Disclosures

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¹ Casey Eye Institute, Oregon Health & Science University, Portland, Oregon.

² Department of Biomedical Engineering, Oregon Health & Science University, Portland, Oregon.

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HUMAN SUBJECTS: Human subjects were included in this study. Participants were enrolled after informed consent in accordance with an Institutional Review Board (Oregon Health & Science University) approved protocol. This study complied with the Declaration of Helsinki and the Health Insurance Portability and Accountability Act.

No animal subjects were used in this study.

Author Contributions:

Conception and design: Zang, Jia

Data collection: Zang

Analysis and interpretation: Zang, Hwang, Jia

Obtained funding: Jia, Hwang

Overall responsibility: Zang, Hormel, Hwang, Jia

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Abbreviations and Acronyms:

3D = 3-dimensional; **AUC** = area under receiver operating characteristic curve; **DCP** = deep capillary plexus; **DME** = diabetic macular edema; **DR** = diabetic retinopathy; **FAZ** = foveal avascular zone; **ICP** = intermediate capillary plexus; **NPA** = nonperfusion area; **OCTA** = OCT angiography; **SVC** = superficial vascular complex; **vtDR** = vision-threatening diabetic retinopathy.

Keywords:

Diabetic retinopathy, OCT angiography, Oxygen distribution, Retinal ischemia.

Correspondence:

Yali Jia, PhD, Casey Eye Institute & Department of Biomedical Engineering, Oregon Health & Science University, 515 SW Campus Dr., CEI 3154, Portland, OR 97239-4197. E-mail: jiaya@ohsu.edu.

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